Boyang Gu

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Research Interests

Reliable and evidence-grounded AI for healthcare; clinical NLP; causal inference from longitudinal electronic health records; verifiable reasoning and scientific workflows.

Education

University of Oxford *Oct. 2026 – Present*
DPhil in Psychiatry. Research focus: language-grounded causal inference from electronic mental health records.

Imperial College London *2023 – 2024*
MSc in Computing (Artificial Intelligence and Machine Learning), **Distinction** (80.61%).

Imperial College London *2019 – 2023*
MSci in Mathematics, **First Class Honours** (81.87%).

Research Experience

Research Intern, MIT CSAIL *2025 – Present*
Research under Prof. Wojciech Matusik, Massachusetts Institute of Technology.

- Developing a typed domain-specific language and AI copilot for protein design and structural biology, with reusable objects and validated scientific operators.
- Integrating structure prediction, docking, relaxation, and simulation backends into reproducible workflows with explicit validation and provenance.

DPhil Research, University of Oxford *2026 – Present*
Investigating how longitudinal mental-health clinical text can support time-anchored extraction of causally relevant variables, confounder identification, and observational treatment-effect studies.

Publications

[1] Zeyu Cao*, Yao Lai, **Boyang Gu***, Cheng Zhang, Pedro Gimenes, Jianqiao Lu, *et al.*, "Scaling laws for mixed quantization," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, Main Conference, Oral, 2026. [Paper].

[2] **Boyang Gu***, Hongjian Zhou*, Bradley Max Segal, Jinge Wu, Zeyu Cao, Hantao Zhong, *et al.*, "Clinical-R1: Empowering large language models for faithful and comprehensive reasoning with clinical objective relative policy optimization," in *Proceedings of the Second AAAI Bridge Program on AI for Medicine and Healthcare*, ser. Proceedings of Machine Learning Research, vol. 317, PMLR, 2026, pp. 117–126. [Paper].

[3] Fenglin Liu*, Hongjian Zhou*, **Boyang Gu***, Xinyu Zou*, Jinfa Huang*, Jinge Wu*, *et al.*, "Application of large language models in medicine," *Nature Reviews Bioengineering*, vol. 3, no. 6, pp. 445–464, 2025. [DOI].

[4] Hongjian Zhou*, **Boyang Gu***, and Chenghao Jin, "Reinforcement learning approach for multi-agent flexible scheduling problems," *Journal of Physics: Conference Series*, vol. 2580, no. 1, p. 012 053, 2023. [DOI].

* Equal contribution (co-first author).

Selected Research Projects

Clinical-R1: Verifiable Reinforcement Learning for Clinical Reasoning *2025 – 2026*

- Developed Clinical-Objective Relative Policy Optimization (CRPO), combining verifiable reward signals for accuracy, faithfulness, and comprehensiveness.
- Trained and evaluated Clinical-R1-3B against standard GRPO approaches. [Code and models]

Hospital-Course Summarization from MIMIC-III *2024*

- Constructed a hospital-course summarization dataset from MIMIC-III and explored supervised fine-tuning (SFT) and direct preference optimization (DPO).
- Developed a rewriting pipeline aimed at reducing factual inconsistency in generated summaries. [Project]

Scaling Laws for Mixed Quantization *2024 – 2026*
Studied how post-training quantization loss varies with model size, quantization ratio Q_r, and quantization block size Q_b; accepted for an EMNLP 2026 oral presentation.

Awards and Teaching

- Dean's List, Imperial College London (2020, 2021).
- Peer Tutor for first-year students, Imperial College London (2022–2023).